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Paper Title : A Comparative Analysis of Hybrid Deep Learning Architectures for Detecting Schizophrenia from EEG Signals



Abu Omayed
Department of Statistics
University of Chittagong
Chittagong, Bangladesh

MD. Asif Iqbal
Department of CSE
CUET
Chittagong, Bangladesh

Muhammad Alamgir Islam
Associate professor
Department of Statistics
University of Chittagong

OUTLINE OF PRESENTATION

- ❑ Introduction
- ❑ Objectives
- ❑ Methodology
- ❑ Proposed Work
- ❑ Result and Discussion
- ❑ Comparison
- ❑ Conclusion
- ❑ Reference

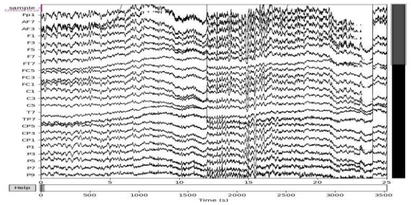
INTRODUCTION

- Schizophrenia is a severe mental disorder that affects thoughts, feelings, and actions.
- Conventional diagnostic methods rely heavily on clinical observation and patient self-reporting, leading to inconsistencies and delays in diagnosis.
- Deep learning offers a promising approach to automatically detect schizophrenia from EEG signals.

OBJECTIVES

- Preprocess EEG data to enhance robustness and generalization capabilities.
- Develop an automated and reliable schizophrenia detection framework from EEG signals using Deep Learning techniques.
- Explore various Hybrid deep learning architectures (e.g., CNN-LSTM, CNN-GRU, CNN-LSTM-GRU) for analysing EEG signals.

METHODOLOGY



EEG dataset

Preprocessing

- Segmentation
- Denoising
- Normalization

DL Models

- 1D-CNNs
- LSTM
- GRU

Hybrid DL Models

- 1D-CNN RNN
- 1D-CNN LSTM
- 1D-CNN GRU
- 1D-CNN SVM
- 1D-CNN LSTM GRU

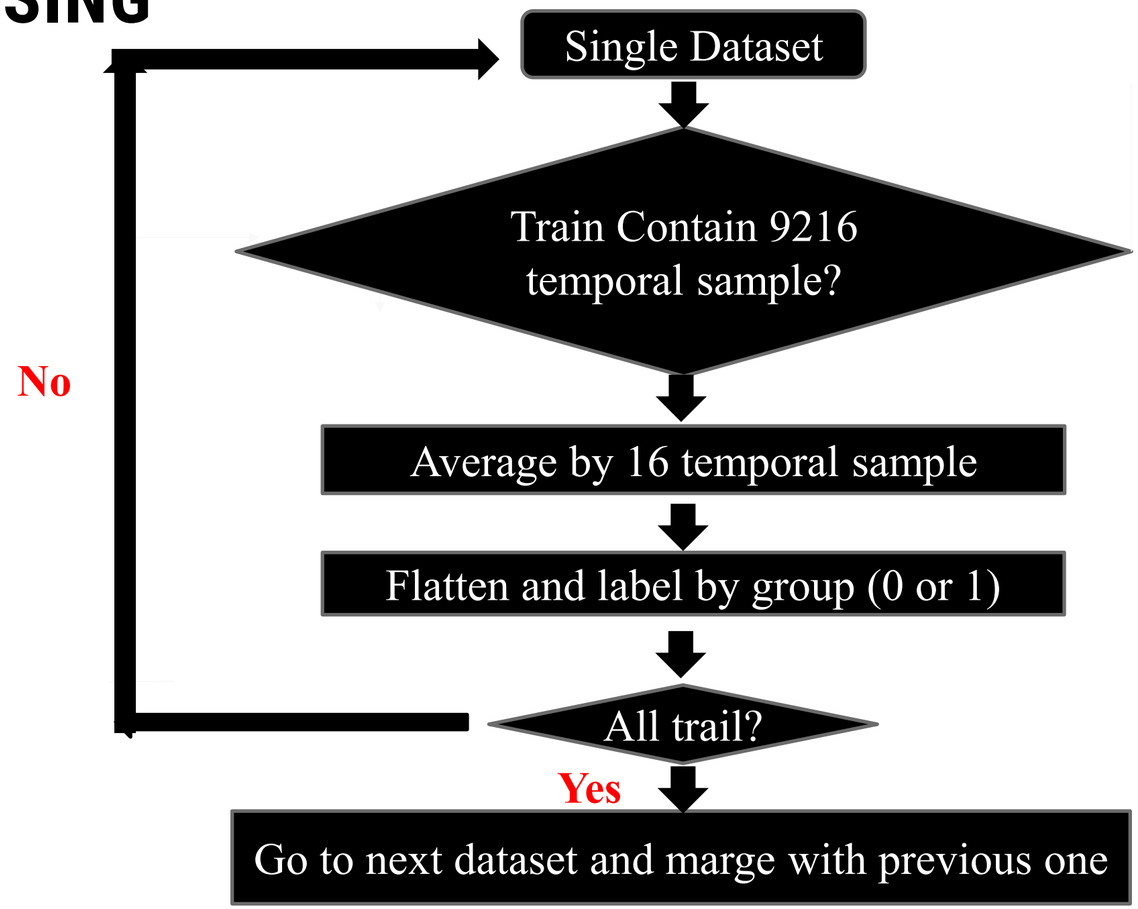
Cross Validation

- Training set
- Test set

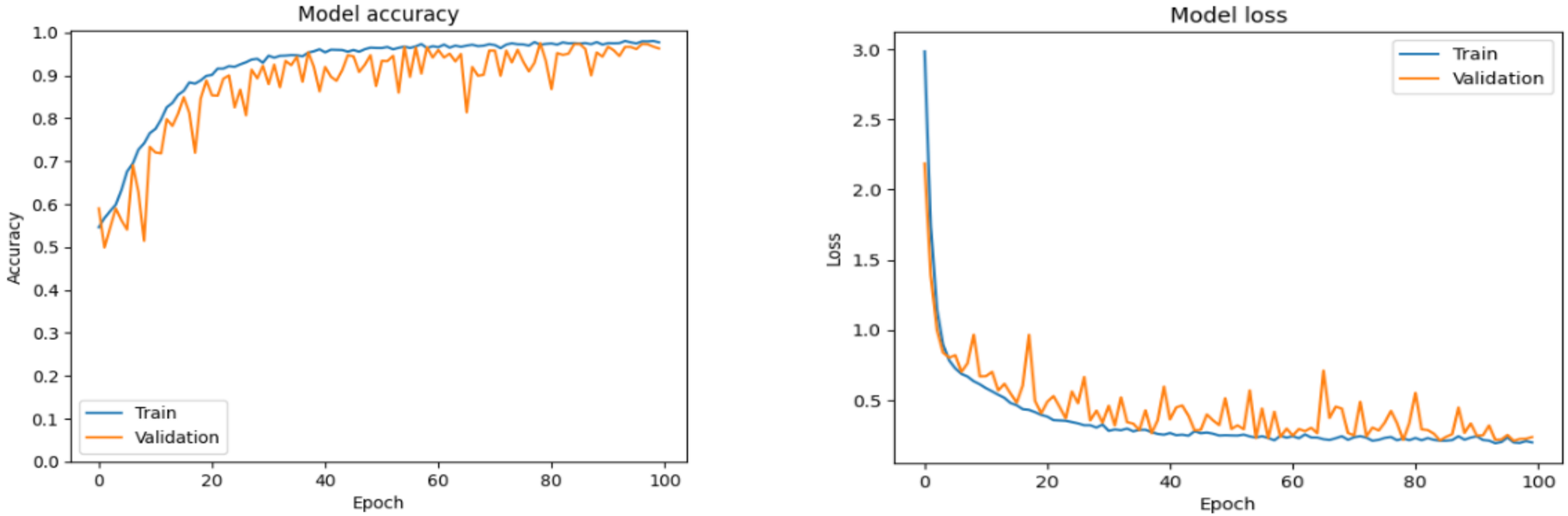
Evaluation

- Accuracy
- Precision
- Recall
- F1-Score

DATA PROCESSING

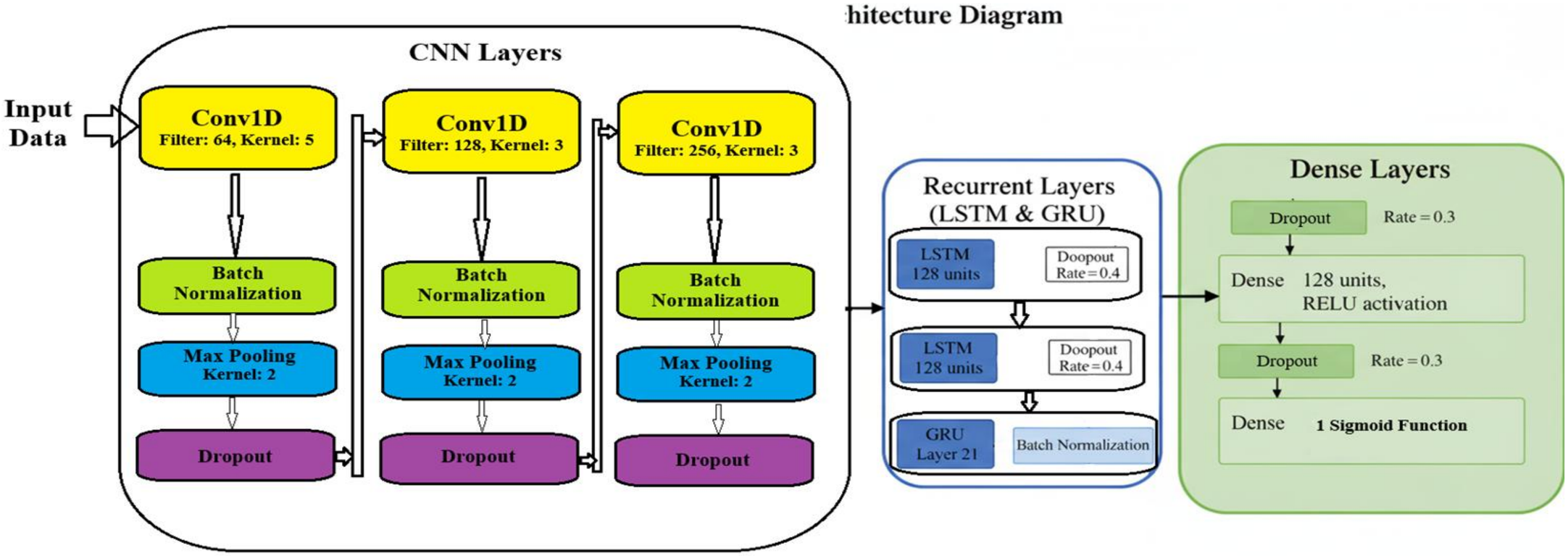


CNN-GRU

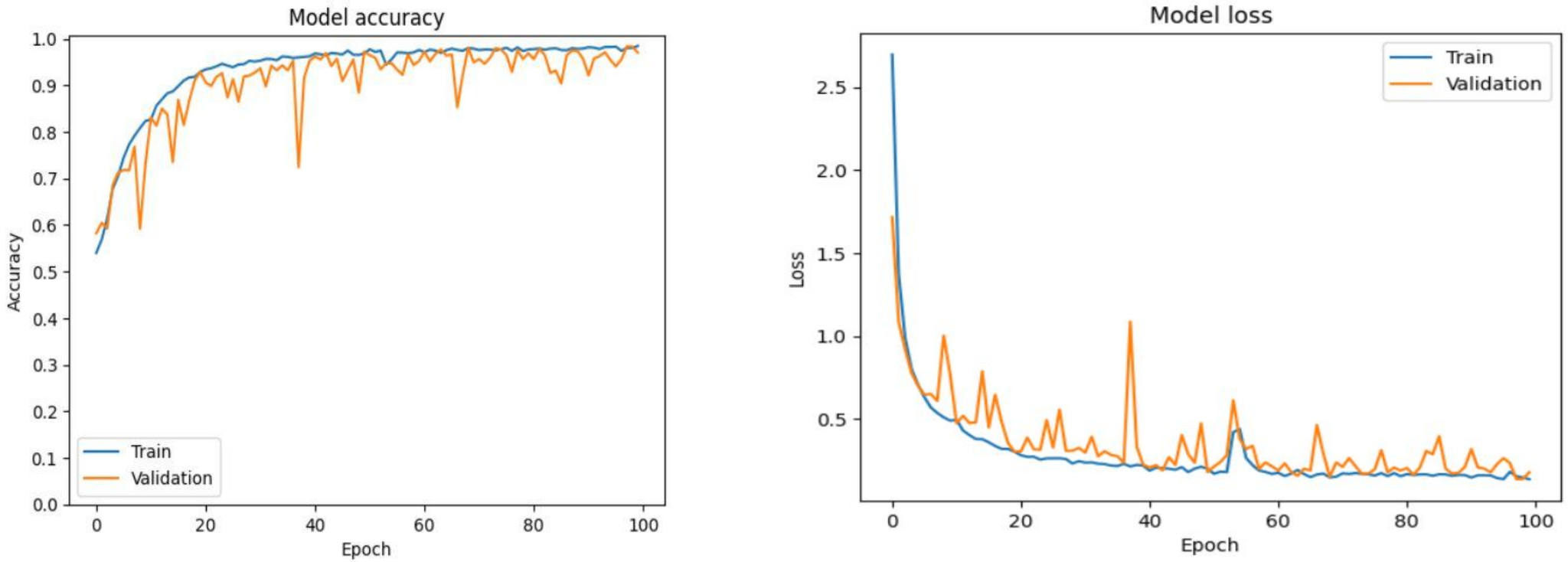


a) Accuracy (b) Loss

CNN-LSTM-GRU Architecture (Proposed Model)

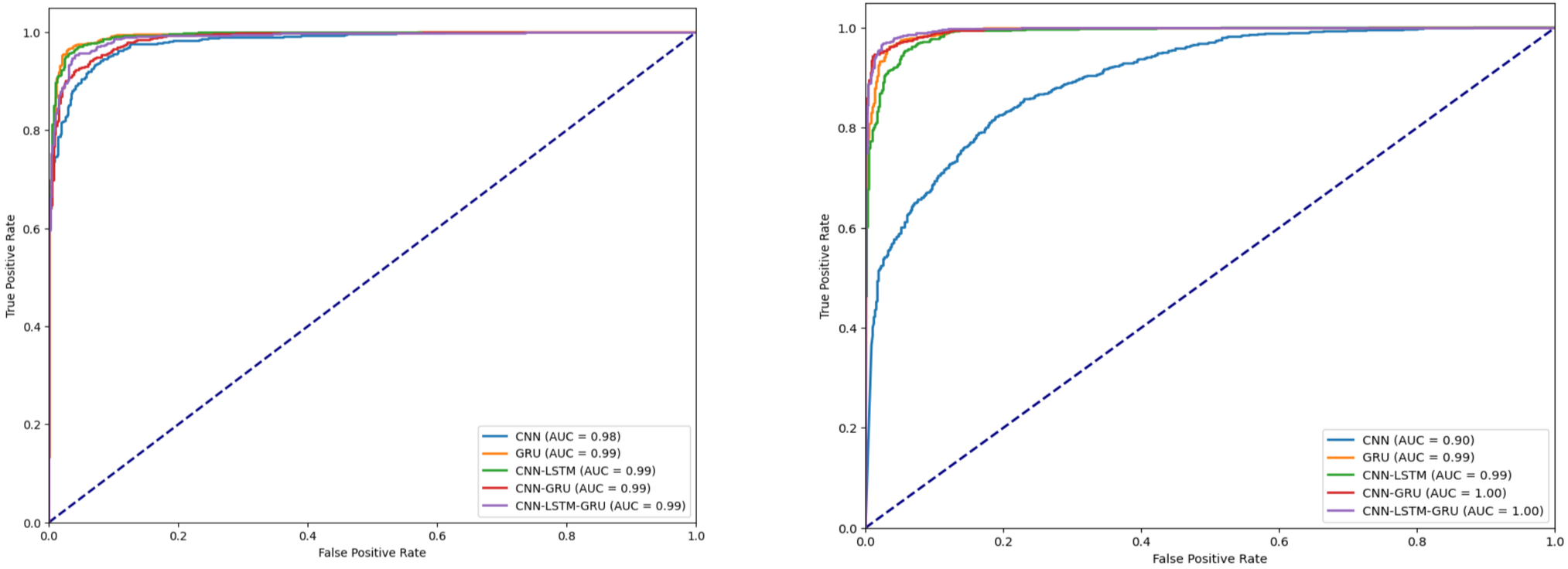


CNN-LSTM-GRU



a) Accuracy (b) Loss

ROC CURVE

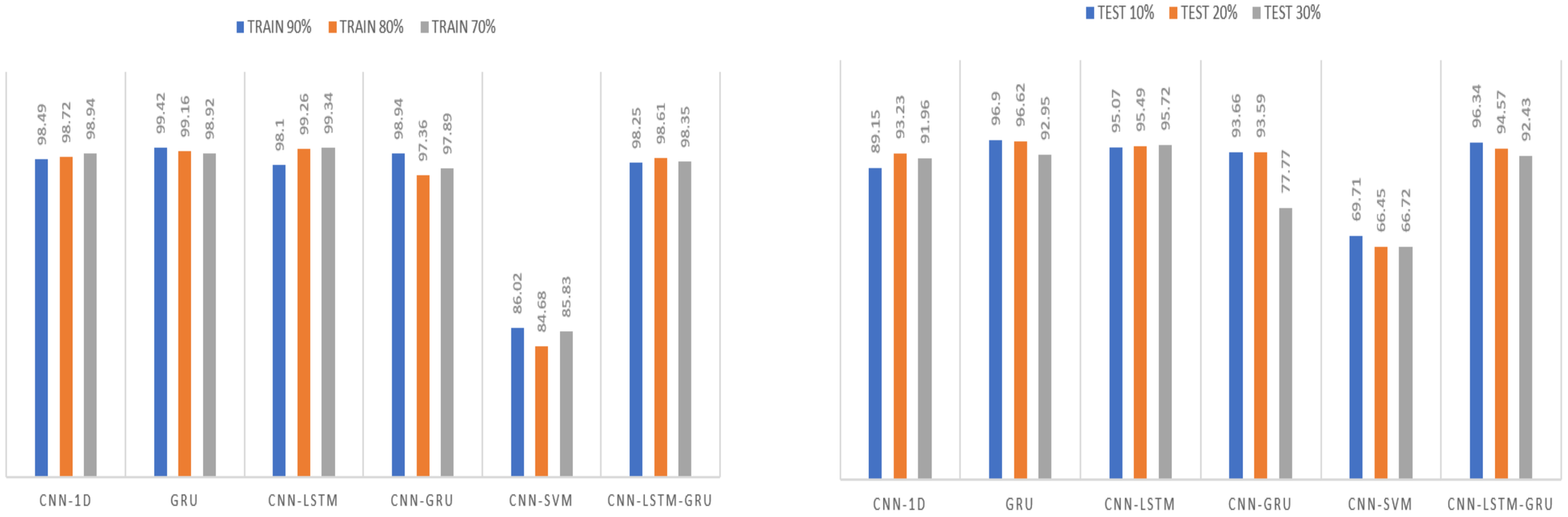


a) ELU (b) ReLu

ACTIVATION FUNCTION COMPARISON

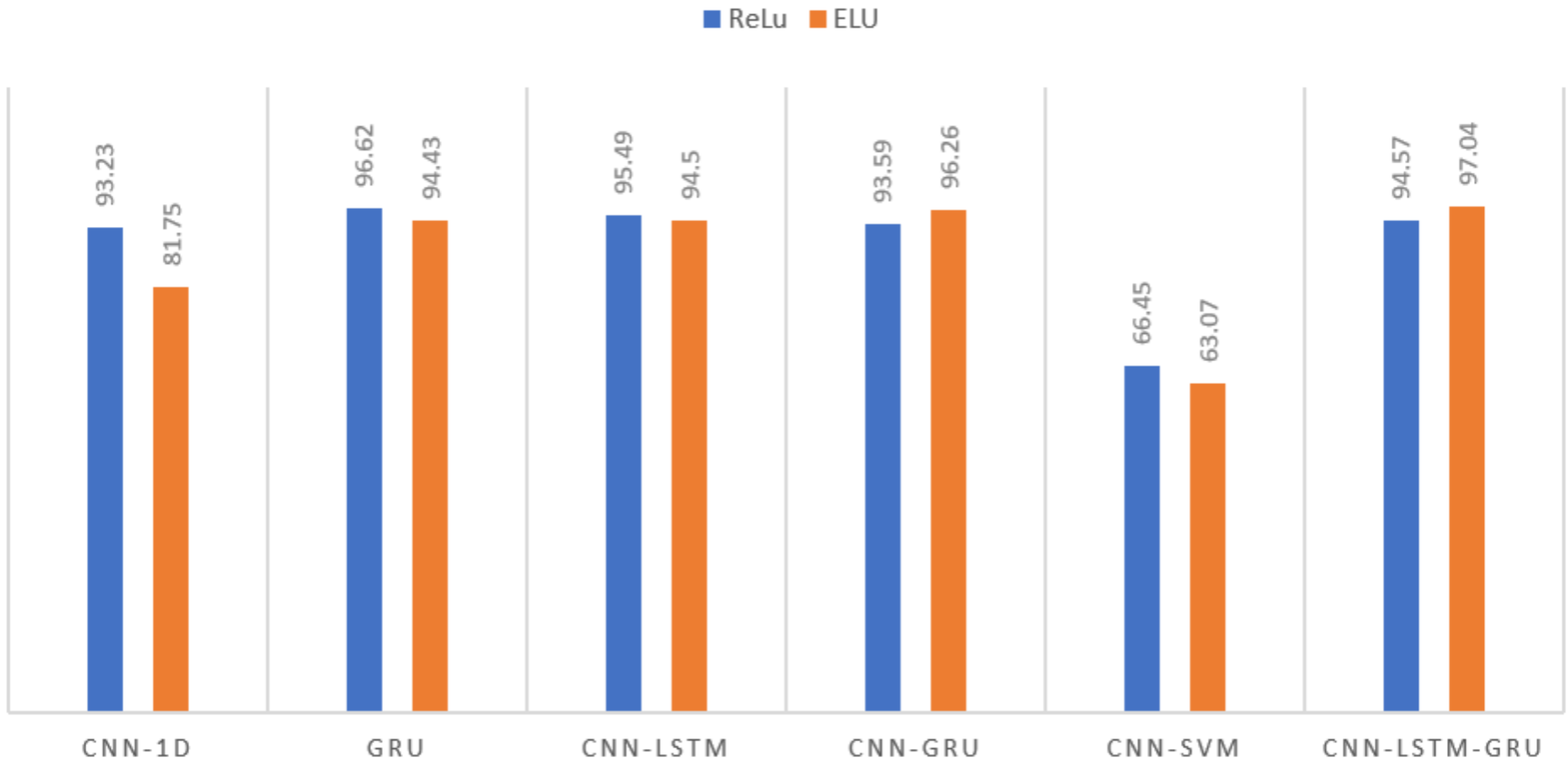
Model	ReLu				ELU			
	Accuracy	Precision	Recall	F1-core	Accuracy	Precision	Recall	F1-score
CNN-1D	93.23	93	93	93	81.75	82	76	82
LSTM	55.11	55	57	60	59.41	97	61	74
GRU	96.62	97	96	97	94.43	93	95	94
CNN-RNN	66.88	69	67	68	59.20	81	63	68
CNN-LSTM	95.49	96	95	95	94.50	93	95	95
CNN-GRU	93.59	94	93	94	96.26	96	96	96
CNN-SVM	66.45	65	64	64	63.07	62	63	62
CNN-LSTM-GRU	94.57	95	94	94	97.04	97	97	97

3-FOLD CROSS VALIDATION



a) Train (b) Test

COMPARISON BETWEEN ReLU AND ELU



COMPARISON

Reference	Year	Subjects	Models	Accuracy
Taylor et al. [1]	2017	21	SVM	80.84%
L. Zang et al. [2]	2019	81	Random Forest	88.00%
Krishnan et al. [3]	2020	14	Various, SVM	93.00%
A. Shoeibi et al. [4]	2021	28	CNN-LSTM	94.75%
Proposed Method	2025	81	CNN_LSTM_GRU	97.04%

CONCLUSION

- This study compares various deep learning and hybrid deep learning algorithms for diagnosing Schizophrenia.
- Our proposed CNN-LSTM-GRU-based triple-hybrid deep learning model achieves the highest accuracy (97.04%) for diagnosis.
- The result demonstrated a novel approach by combining deep learning models to generate better accuracy in analysing EEG signals to diagnose Schizophrenia.

REFERENCE

[1] Taylor, Jeremy A., Natasha Matthews, Patricia T. Michie, Maria J.Rosa, and Marta I. Garrido. ”Auditory prediction errors as individual biomarkers of schizophrenia.” Neuro Image: Clinical 15 (2017): 264-273.

[2] Zhang, L. (2019, July). EEG signals classification using machine learning for the identification and diagnosis of schizophrenia. In 2019 41stannual international conference of the ieee engineering in medicine and biology society (EMBC) (pp. 4521-4524). IEEE.

[3] Krishnan, Palani Thanaraj, Alex Noel Joseph Raj, Parvathavarthini Balasubramanian, and Yuanzhu Chen. ”Schizophrenia detection using Multivariate Empirical Mode Decomposition and entropy measures from multi channel EEG signal.” Biocybernetics and Biomedical Engineering40, no. 3 (2020): 1124-1139.

[4] Shoeibi, Afshin, Delaram Sadeghi, Parisa Moridian, Navid Ghassemi, Jonathan Heras, Roohallah Alizadehsani, Ali Khadem et al. ”Automatic diagnosis of schizophrenia in EEG signals using CNN-LSTM models.” Frontiers in neuroinformatics 15 (2021): 777977.

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Thank You!